

AP Calculus BC: The Complete Exam Guide

All 10 units explained, including series, parametric and polar, a free-response playbook with justification language, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3:10**

hours total
100 + 90 minutes

42+6

multiple-choice +
free-response questions

2/3

of the exam is
no-calculator

79–82%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, four parts

Your AP score (1–5) combines multiple choice and free response, each worth half. Both sections are split into a no-calculator part and a calculator part, and most of the exam is done without one.

35%

MC Part A • 29 Q • no calc

15%

MC Part B • 13 Q • ...

17%

FR Part A • calc

33%

FR Part B • 4 Q • no calc

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 40 min 42 questions	Part A: 29 questions, no calculator (35% of score). Part B: 13 questions, graphing calculator required for some (15%). Functions appear as equations, graphs, tables and words.
II • Free response	50% of score 1 hr 30 min 6 questions	Part A: 2 questions, graphing calculator required (about 17%). Part B: 4 questions, no calculator (about 33%). At least two questions use a real-world context.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

CALCULATOR RULES

Know your four calculator skills

On calculator parts you are expected to **graph** a function, **solve** an equation, find a **numerical derivative** and evaluate a **definite integral**. Check that your model is on College Board's approved list.

How to use this guide

Just starting?

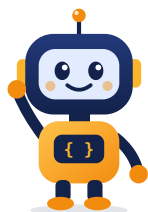
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 25 minutes, then use the final-four-weeks plan.



Pigo's tip: Your score report also includes an **AB subscore** based on the AB topics in the exam, so strong AB skills protect your result.

SECTION I

02 The multiple-choice section

Part A gives you 29 questions in 62 minutes without a calculator (about 2 minutes each). Part B gives 13 questions in 38 minutes with a calculator (about 3 minutes each).

29

no-calculator questions
in Part A

13

calculator-allowed
questions in Part B

~2 min

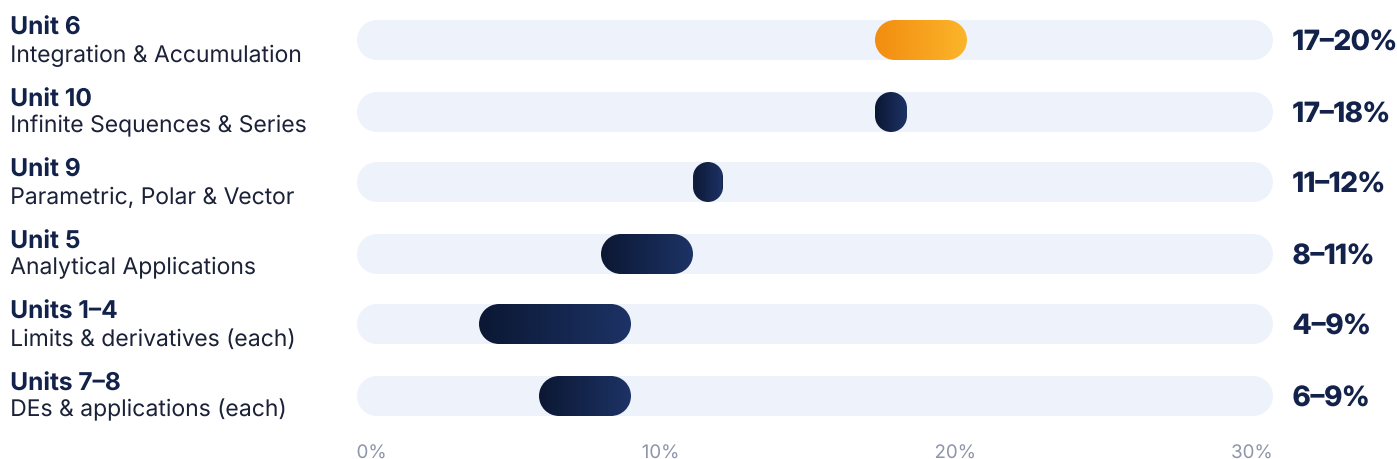
per question in Part A

0

points lost for a wrong
answer, so never leave
one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.



What this means for your study time: Series (Unit 10) is as heavy as integration and is the topic BC students most often under-prepare. Start it early and revisit it every week.

Strategies that work on this section

- Read graphs of f' carefully. Many questions show the derivative, not the function. Increasing f means $f' > 0$, not f' increasing.
- Know your limits toolkit: factoring, conjugates, the special trig limits and L'Hospital's Rule.
- Units tell you the answer type. A rate in gallons per hour, integrated over hours, gives gallons.
- In Part B, use the calculator for definite integrals and solving. Store intermediate values, don't round.
- Eliminate with estimates. A quick sketch often rules out two choices.
- Flag and move on after 3 minutes. Come back once the quick points are banked.

03 WHAT THE COURSE COVERS

The 10 units, in plain English

BC covers everything in AB plus two more units and a few extra topics.

UNIT 1 4–7%	Limits & Continuity Be ready to: evaluate limits from graphs, tables and algebra; find asymptotes; apply continuity and the IVT.	UNIT 2 4–7%	Differentiation: Definition & Rules Be ready to: use the limit definition and all basic derivative rules; connect differentiability and continuity.
UNIT 3 4–7%	Composite, Implicit & Inverse Be ready to: apply the chain rule, implicit differentiation and inverse-function derivatives.	UNIT 4 6–9%	Contextual Applications Be ready to: interpret rates with units; solve motion and related-rates problems; linearize; use L'Hospital's Rule.
UNIT 5 8–11%	Analytical Applications Be ready to: apply MVT and EVT; justify extrema and concavity; solve optimization problems.	UNIT 6 17–20%	Integration & Accumulation Be ready to: use Riemann sums and the FTC; integrate by substitution, parts and partial fractions ; evaluate improper integrals .
UNIT 7 6–9%	Differential Equations Be ready to: read slope fields; use Euler's method ; solve separable equations; analyze logistic growth .	UNIT 8 6–9%	Applications of Integration Be ready to: find average value, area, volume and arc length of a curve.
UNIT 9 11–12%	Parametric, Polar & Vector Be ready to: find derivatives, speed and arc length of parametric curves; analyze vector-valued motion; find polar slopes and areas.	UNIT 10 17–18%	Infinite Sequences & Series Be ready to: apply the convergence tests; find radius and interval of convergence; build Taylor and Maclaurin series; bound errors with the alternating series and Lagrange bounds.

Bold topics are BC-only. Everything else also appears on AB, and those AB topics produce your AB subscore.

SECTION II

04 The 6 free-response questions

90 minutes, 6 questions, 9 points each. Part A (2 questions, calculator) gets 30 minutes; Part B (4 questions, no calculator) gets 60 minutes.

Q1 · calc	Q2 · calc	Q3	Q4	Q5	Q6
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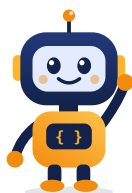
6 questions × 9 points = 54 raw points, worth 50% of your AP score

Rate Rates & accumulation Often Part A, calculator Rates in and out, amounts over time, and maximums. Shared with AB. Amount = start + $\int(\text{in} - \text{out})$.	Param Parametric / vector motion Usually Part A A particle moving in the plane: velocity vector, speed, total distance traveled and position at a time. Speed = $\sqrt{(dx/dt)^2 + (dy/dt)^2}$.	Graph Graph of f' / tables Analysis + Riemann sums Analyze a function from its derivative's graph or from data in a table, with full justifications. Justify with f' and f'' signs.
Polar Polar curves Area and rates Area inside polar curves, between curves, and how r or position changes with θ. Area = $\frac{1}{2} \int r^2 d\theta$. Find the bounds first.	DE Differential equations Euler, slope fields, logistic Slope fields, Euler's method, separable equations and logistic models. Logistic growth is fastest at half the carrying capacity.	Series Taylor series Almost always Q6 Build a Taylor polynomial, find the interval of convergence, and bound the error. Check both endpoints separately with a named test.

◆ A time plan for the 90 minutes

Part A Q1 15 min	Part A Q2 15 min	Q3 15 min	Q4 15 min	Q5 15 min	Q6 15 min
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About 15 minutes per question. In Part B you may return to Part A questions, but without a calculator.



Write the **setup** before you compute. An integral written correctly with limits earns points even if you run out of time.

05

FREE-RESPONSE PLAYBOOK

Anatomy of a full-credit answer

Series questions reward precise language. Here is a short practice prompt with a full-credit answer.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Q6 style): Find the interval of convergence of the power series $\sum_{n=1}^{\infty} (x - 2)^n / (n \cdot 3^n)$. Show the work that leads to your answer.

Full-credit response

- ① Ratio test: $|a_{n+1}/a_n| = |x - 2| \cdot n / (3(n + 1)) \rightarrow |x - 2|/3$ as $n \rightarrow \infty$.
- ② The series converges when $|x - 2|/3 < 1$, so $-1 < x < 5$.
- ③ At $x = 5$: $\sum 1/n$ is the harmonic series, which **diverges**.
- ④ At $x = -1$: $\sum (-1)^n/n$ converges by the alternating series test (terms decrease to 0).
- ⑤ Interval of convergence: $-1 \leq x < 5$.

- ① **Set up the ratio test** with absolute values and take the limit.
- ② **Solve the inequality** for the open interval.
- ③ **Test each endpoint** and name the test you use.
- ④ **Give the reason** the alternating series converges.
- ⑤ **Write the final interval** with correct brackets.

Five habits for every free-response question

1

Label every part: (a), (b), (c).

2

Justify with f' or f'' signs, not words like "the graph".

3

Include units in every contextual answer.

4

Give decimals to 3 places on calculator parts.

5

Stuck? Write the setup integral or equation. It still scores.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Forgot the **+C**? I'll take that point!



01 Skipping endpoint checks

The ratio test says nothing at the endpoints. Test each one separately and name the test.

02 Forgetting factorials

The n th Taylor coefficient is $f^{(n)}(a)/n!$. Missing the $n!$ is the most common series slip.

03 Polar bounds

For area, find where the curve starts and ends (or where curves intersect) before setting up $\frac{1}{2}r^2 d\theta$.

04 Speed vs velocity

Speed is the magnitude of the velocity vector. Total distance integrates speed, not dy/dt .

05 d^2y/dx^2 for parametric curves

Differentiate dy/dx with respect to t , then divide by dx/dt again.

06 Justifying with "the graph"

Readers need a reason based on f' or f'' , such as " f' changes from $+$ to $-$ ".

07 Dropping **+C** or the initial condition

Add $+C$ right after integrating and solve for it before isolating y .

08 Rounding too early

Store calculator values and round only the final answer to 3 decimal places.

09 Wrong error bound

Alternating series: error $\leq$ first omitted term. Otherwise use the Lagrange bound with a maximum of the next derivative.

10 Leaving a part blank

Each part is scored separately. A correct setup can earn points even if the next step fails.

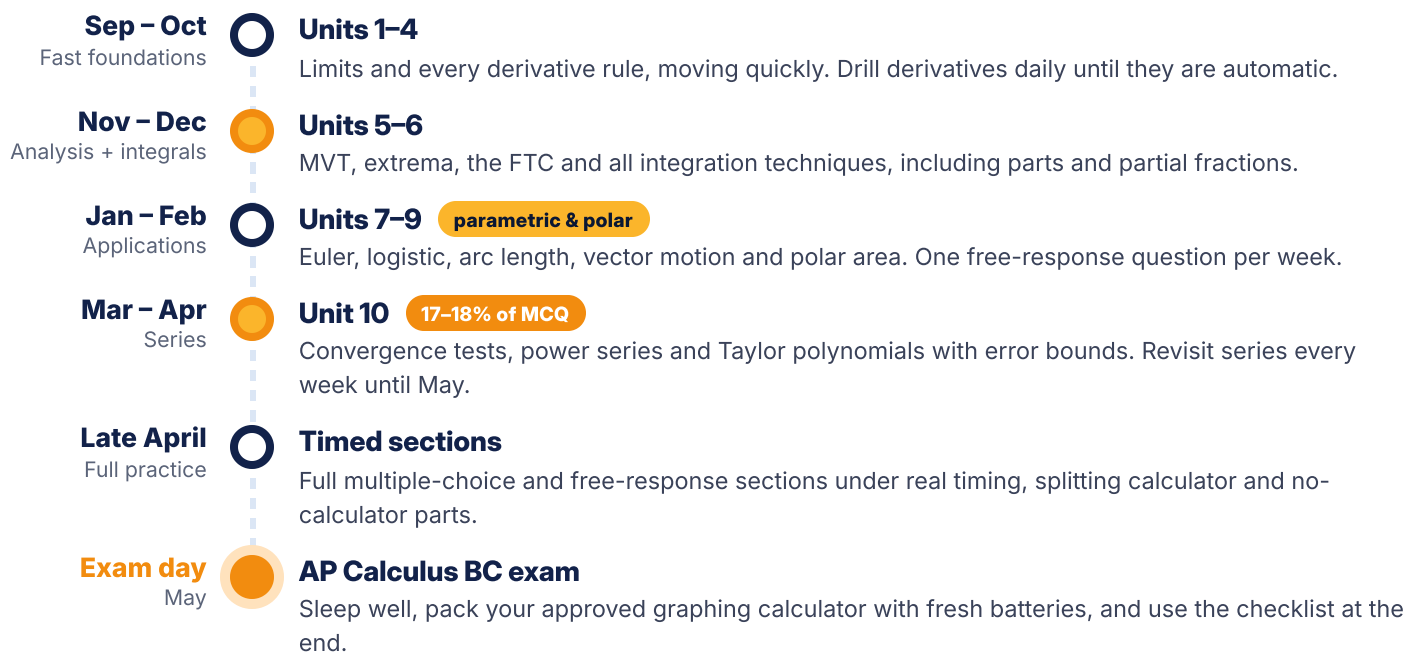
Quick self-check: if $f'(2) = 0$ and $f''(2) = -3$, what can you conclude about f at $x = 2$? (Answer: a relative maximum, by the second derivative test.)



PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep reviewing old units as you learn new ones.



◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: one full no-calculator Part A and two free-response questions	List your three weakest units
3 weeks out	Drill weak units; one free-response question of each type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish every part on time
Final week	Light review: derivative/integral rules, justification phrases	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. No calculator unless marked. Give yourself **25 minutes** for all ten, then check page 10.

1 UNIT 10 • CONVERGENCE

Which of the series converge **conditionally**?

I. $\sum (-1)^n/\sqrt{n}$ II. $\sum (-1)^n/n^2$ III. $\sum (-1)^n \cdot n/(n+1)$

- A II only
B I and II
C I and III
D I only

2 UNIT 10 • MACLAURIN SERIES

What is the coefficient of x^3 in the Maclaurin series for $f(x) = e^x \sin x$?

- A $1/6$
B $1/3$
C $1/2$
D $2/3$

3 UNIT 10 • INTERVAL OF CONVERGENCE

What is the interval of convergence of $\sum_{n=1}^{\infty} (x+1)^n / (n \cdot 2^n)$?

- A $-3 \leq x < 1$
B $-3 < x < 1$
C $-3 < x \leq 1$
D $-3 \leq x \leq 1$

4 UNIT 9 • PARAMETRIC

A curve is given by $x = t^2$, $y = t^3 - 3t$. What is d^2y/dx^2 at $t = 1$?

- A 0
B $3/4$
C $3/2$
D 3

5 UNIT 9 • POLAR AREA

What is the area enclosed by the polar curve $r = 2 + 2 \cos \theta$?

- A 3π
B 4π
C 6π
D 8π

6 UNIT 9 • ARC LENGTH

What is the length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \leq t \leq 1$?

- A $2\sqrt{2}$
B $4\sqrt{2} - 2$
C $2\sqrt{2} - 1$
D 6

7 UNIT 6 • INTEGRATION BY PARTS

What is $\int_0^1 x^2 e^x dx$?

- A $e - 1$
B $2e - 1$
C $e/2$
D $e - 2$

8 UNIT 6 • IMPROPER INTEGRAL

What is the value of $\int_2^{\infty} 1/(x^2 - 1) dx$?

- A $(\ln 3)/2$
B $\ln 3$
C $1/2$
D The integral diverges

9 UNIT 7 • EULER'S METHOD

$dy/dx = x + y$ with $y(0) = 1$. Using Euler's method with two steps of size 0.5, what is the estimate for $y(1)$?

- A 2
B 2.25
C 2.5
D 3

10 UNIT 7 • LOGISTIC GROWTH

A population satisfies $dP/dt = 0.4P(1 - P/500)$ with $P(0) = 100$. At what population is it growing fastest?

- A 100
B 200
C 250
D 500

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	D - I only	I converges by the alternating series test but $\sum 1/\sqrt{n}$ diverges, so it is conditional. II converges absolutely. III diverges because its terms don't approach 0.
2	B - $1/3$	$e^x = 1 + x + x^2/2 + \dots$, $\sin x = x - x^3/6 + \dots$. The x^3 terms: $x^2/2 \cdot x - x^3/6 = (1/2 - 1/6)x^3 = x^3/3$.
3	A	Ratio test gives $ x + 1 < 2$. At $x = 1$ the harmonic series diverges; at $x = -3$ the alternating harmonic series converges.
4	C - $3/2$	$dy/dx = (3t^2 - 3)/(2t) = 3t/2 - 3/(2t)$. Its t-derivative is $3/2 + 3/(2t^2) = 3$ at $t = 1$. Divide by $dx/dt = 2$: $3/2$.
5	C - 6π	$\frac{1}{2} \int_0^{2\pi} (2 + 2\cos \theta)^2 d\theta = \frac{1}{2} \int_0^{2\pi} (4 + 8\cos \theta + 4\cos^2 \theta) d\theta = \frac{1}{2} (8\pi + 4\pi) = 6\pi$.
6	B - $4\sqrt{2} - 2$	Speed = $\sqrt{(36t^2 + 36t^4)} = 6t\sqrt{1 + t^2}$. $\int_0^1 6t\sqrt{1 + t^2} dt = 2(1 + t^2)^{3/2}$ from 0 to 1 = $4\sqrt{2} - 2$.
7	D - $e - 2$	Parts twice: $\int x^2 e^x dx = e^x(x^2 - 2x + 2)$. From 0 to 1: $e(1) - 2 = e - 2$.
8	A - $(\ln 3)/2$	Partial fractions: $1/(x^2 - 1) = \frac{1}{2}[1/(x - 1) - 1/(x + 1)]$. Antiderivative $\frac{1}{2} \ln((x - 1)/(x + 1)) \rightarrow 0$ at ∞ and $\frac{1}{2} \ln(1/3)$ at 2, giving $(\ln 3)/2$.
9	C - 2.5	Step 1: slope $0 + 1 = 1$, so $y(0.5) \approx 1.5$. Step 2: slope $0.5 + 1.5 = 2$, so $y(1) \approx 1.5 + 0.5(2) = 2.5$.
10	C - 250	Logistic growth is fastest at half the carrying capacity: $500/2 = 250$. The initial value doesn't change this.

9-10

Strong. Move on to full timed free-response practice.

6-8

On track. Revisit the unit behind every miss on page 4.

0-5

Build foundations now. Drill integration techniques and series tests (Units 6 and 10) weekly.

10 KNOW THE TARGET

How students score

In recent years, about 79–82% of students earned a 3 or higher on AP Calculus BC, and about 44–46% earned a 5. Steady preparation is what moves a score up a full point.

79–82%

of students earned a 3 or higher in recent years

44–46%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved graphing calculator** with fresh batteries, in radian mode
- ☐ Plan your time: about **2 min** per Part A question and **15 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Give calculator answers to **3 decimal places**

In the free-response section: label each part, show the setup before computing, justify with derivative signs or theorem conditions, and include units in every contextual answer.



ACT SAT GO • 1-ON-1 AP TUTORING

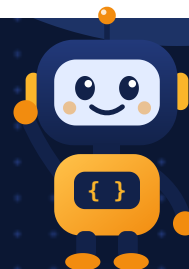
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Book a free diagnostic lesson. We'll find your weakest units, review your practice responses, and give you a week-by-week plan to exam day in May.

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